Year 9 Mathematics

Unit 1: Number

1.3 Indices

Indices

STARTER · 5 MIN

Date: _____ **Title: Indices**

LEARNING OBJECTIVES

- Use the laws of indices
- Use zero and negative indices
- Work out powers of fractions

STARTER *Answer in your book. No calculators.*

1. Work out 5.93×10
2. Work out $482 \div 10$
3. Work out 2^4
4. Work out 3^3

Starter — Answers

STARTER

Tick or correct in green pen

1. **59.3**
2. **48.2**
3. **16**
4. **27**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the laws of indices	☐
SILVER	I can use zero and negative indices	☐
GOLD	I can work out powers of fractions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Indices

INTRODUCTION · 3 MIN

In 5^3, 5 is the base and 3 is the index (power).

Multiply

Add the indices: $a^m \times a^n = a^{m+n}$

$2^3 \times 2^4 = 2^7$

Divide

Subtract the indices: $a^m \div a^n = a^{m-n}$

$2^6 \div 2^2 = 2^4$

Power of a power

Multiply the indices: $(a^m)^n = a^{mn}$

$(2^3)^2 = 2^6$

Zero and negative

$2^0 = 1$, and a negative index means 1 over.

$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

Examples 1 and 2 — Laws of Indices

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$

Example 1

Write as a single power.

$7^3 \times 7^5 \div 7^2$

Example 2

Write as a single power.

$(2^4)^3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Add, then subtract

$7^3 \times 7^5 = 7^8$

$7^8 \div 7^2 = 7^6$

Tip: 3 + 5 − 2 = 6

Example 2

Multiply the indices

$2^4 \times 2^4 \times 2^4$

$= 2^{12}$

Tip: not 2^7: three lots of 4.

Task 1

PRACTICE · 9 MIN

BRONZE *Write as a single power*

Hint $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$

1 $3^4 \times 3^5$

2 $7^9 \div 7^5$

3 $(5^5)^2$

4 $6^2 \times 6^7 \div 6^3$

5 $(4^3)^4$

★ $(2^3)^2 \times 2^4$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 3^{9}

2 7^{4}

3 5^{10}

4 6^{6}

5 4^{12}

★ 2^{10}

Examples 3 and 4 — Zero and Negative Indices

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint

$a^0 = 1$ $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $a^{-n} = 1 \div a^n$

Example 3

Work out the value of

(a) 5^0 (b) 2^{-3}

Example 4

Write as a single power.

(a) $7^9 \div 7^{-5}$ (b) $(6^{-2})^{-3}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Power 0 is 1; negative: 1 over

(a) $5^0 = 1$

(b) $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

Tip: a negative index is not negative.

Example 4

Same laws with negative indices

(a) $9 - (-5) = 14$, so 7^{14}

(b) $-2 \times -3 = 6$, so 6^6

Tip: $- \times - = +$, so not 6^{-5}.

Task 2

PRACTICE · 9 MIN

SILVER

Work out, or write as one power

Hint $a^0 = 1$ $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$ $a^{-n} = 1 \div a^n$

1 5^0

2 4^{-1}

3 3^{-2}

4 $7^8 \times 7^{-3}$

5 $2^{-4} \div 2^{-7}$

★ $(5^{-3})^{-2}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 1

2. $\frac{1}{4}$

3. $\frac{1}{9}$

4. 7^5

5. 2^3

★ 5^6

Examples 5 and 6 — Powers of Fractions

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$

Example 5

(3 marks)

Work out

(a) $\left(\frac{2}{3}\right)^3$ (b) $\left(1\frac{1}{2}\right)^2$

Example 6

(2 marks)

Work out

$\left(\frac{3}{4}\right)^{-2}$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Power of top and bottom

(a) $\frac{2^3}{3^3} = \frac{8}{27}$

(b) $\left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2\frac{1}{4}$

Tip: change a mixed number to a fraction first.

Example 6

Negative: flip the fraction

$\left(\frac{4}{3}\right)^2 = \frac{4^2}{3^2}$

$= \frac{16}{9}$

Tip: then square top and bottom.

Task 3

PRACTICE · 9 MIN

GOLD *Work out the value*

Hint $(\frac{a}{b})^n = \frac{a^n}{b^n}$ $(\frac{a}{b})^{-n} = (\frac{b}{a})^n$

1 $(\frac{3}{4})^2$

2 $(\frac{2}{5})^3$

3 $(1\frac{1}{3})^2$

4 $(\frac{1}{3})^{-2}$

5 $(\frac{2}{3})^{-1}$

★ $(\frac{1}{4})^{-3}$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $\frac{9}{16}$

2 $\frac{8}{125}$

3 $\frac{16}{9} = 1\frac{7}{9}$

4 9

5 $\frac{3}{2}$

★ 64

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write $6^4 \times 6^3 \div 6^5$ as a single power.

SILVER Work out the value of 5^{-2}

GOLD Work out $\left(\frac{3}{5}\right)^2$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE 6^2

SILVER $\frac{1}{25}$

GOLD $\frac{9}{25}$

Exam Practice — Laws of Indices

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$

Question 1

(2 marks)

Write as a power of 5.

(a) $5^8 \times 5^3$ (1)

.....

(b) $(5^2)^4$ (1)

.....

Question 2

(2 marks)

Write $(7^3)^2 \div 7^4$ as a single power of 7.

.....

Exam Practice — Zero and Negative Indices

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint

$a^0 = 1$ $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $a^{-n} = 1 \div a^n$

Question 3

(2 marks)

Work out the value of

(a) 8^0 (1)

.....

(b) 4^{-2} (1)

.....

Question 4

(1 mark)

Maisa writes $(6^{-2})^{-3} = 6^{-5}$

Is she correct? Explain your answer.

.....

Exam Practice — Powers of Fractions

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $(\frac{a}{b})^n = \frac{a^n}{b^n}$ $(\frac{a}{b})^{-n} = (\frac{b}{a})^n$

Question 5

(2 marks)

Work out the value of

$(2\frac{1}{2})^2$

.....

Question 6

(2 marks)

Samia writes

$(\frac{2}{3})^{-2} = \frac{4}{9}$

Is she correct? Explain your answer.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 5^{11} — B1

(b) 5^{8} — B1

Tip: multiply: add; power of a power: multiply

Question 2

$(7^3)^2 = 7^6$ — M1

$7^6 \div 7^4 = 7^2$ — A1

Tip: multiply, then subtract the indices.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 1 — B1

(b) $\frac{1}{16}$ — B1

Tip: $4^{-2} = 1 \div 4^{2}$

Question 4

No: $-2 \times -3 = 6$, so 6^{6} — B1

Tip: power of a power: multiply.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$\left(\frac{5}{2}\right)^2 = \frac{25}{4}$ M1

$\mathbf{\frac{25}{4} = 6\frac{1}{4}}$ A1

Tip: change to a fraction before squaring.

Question 6

$\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ M1

No, it is $\frac{9}{4}$ A1

Tip: a negative index flips the fraction.